

DESIGN AND DEVELOPMENT OF FUEL AND FUEL ELEMENTS FOR A PROTOTYPE FAST REACTOR

J. F. W. BISHOP,* C. BOORMAN,* G. O. JACKSON,* D. L. LINNING,*
N. PARKINSON† and A. E. WILLIAMS‡

1. Introduction

Uranium-plutonium ceramic fuel was chosen for the PFR in 1962. At that date, evidence from the U.S.S.R. reactor BR 5 and from U.S. and U.K. experimental programmes suggested that the burnup and temperature requirements of an economic sodium cooled power reactor were reasonable development targets with ceramic fuel for a start-up date around 1970. Ceramic fuel was chosen in place of metal on the grounds that the development problems were regarded as less difficult with the former, that very large thermal reactor development programmes were studying ceramic fuel, and that more development potential in terms of varying reactor parameters appeared to be available to ceramic fuel. The same view is still taken, but U.K. opinion is that the choice is not obvious, and that a metal fuelled system is a viable target.

This paper considers the design requirements of a sodium cooled fast reactor ceramic fuel element and reports relevant data which have accrued on the U.K. programme and the progress made on the design and development of the PFR fuel. Conference papers by Lawton *et al.*⁽¹⁾ and Nettley *et al.*⁽²⁾ are essentially complementary to this paper.

2. Fuel and Cladding Characteristics

The PFR development studies have been restricted to two fuels—mixed oxide and mixed carbide—and two types of canning materials—austenitic stainless steels of the 316 type and precipitation hardening nickel base alloys, principally PE 16 (see Nettley *et al.*⁽²⁾).

* UKAEA, Risley.

† UKAEA, Dounreay.

‡ UKAEA, Springfields.

Fuel

The main fuel performance characteristics relevant to fuel element behaviour are release of fission gas, swelling due to fission products, compatibility with cladding, migration of condensible fission products, plutonium migration, and fuel redistribution.

Gas release

Data on gas release from co-precipitated mixed oxide compacted powder fuel and from sintered solid pellets irradiated in a thermal flux to high burnup and at high linear pin rating have been published by General Electric workers.⁽³⁾ At values of $\int k d\theta$ taken between the limits of the fuel surface and centre temperatures (subsequently called $K(\theta)$, where θ is the local cladding temperature) between 20 W/cm and 60 W/cm, maximum clad temperatures between 380°C and 680°C, and burnups up to 12% of heavy atoms, they found that pellet fuel released an average of about 36% of the fission gas, and powder fuel about 47%, with values in the range 10% to 65%. U.K. workers⁽¹⁾ have reported results for thermal and fast flux irradiations of mixed oxide fuel at $K(\sim 500\text{--}700^\circ\text{C})$ values of 14 W/cm and 66 W/cm up to 7.6% (av.) burnup for annular pellets and vibro-compacted granule fuel, all of about 80% density in terms of the fuelled portion of the pins. They observe that the release from annular pellets is significantly less than from vibro fuel. For vibro fuel with $K(650^\circ\text{C})$ of about 30 W/cm, they predict a release of about 90% at 10% burnup, with annular pellet fuel significantly lower at about 50%. The data on gas release from carbide fuel are more sparse. Strasser *et al.*⁽⁴⁾ have reported results on $(\text{U}_{0.8}\text{Pu}_{0.2})\text{C}$ of hyperstoichiometric carbon equivalence up to nearly 12% (max.) burnup at average $K(\sim 550^\circ\text{C})$ values of about 43 W/cm. Gas release up to 5% burnup is only 1% or 2% in solid pellet fuel, rising to up to around 30% at about 10% burnup. Values as high as 21% have been reported by ANL workers⁽⁵⁾ in physically mixed strongly hypostoichiometric PuC in 80 wt.% stoichiometric UC at 2% burnup with $K(400^\circ\text{C})$ values of 22 W/cm. U.K. results up to 5% burnup and $K(650^\circ\text{C})$ values of about 30 W/cm show a few per cent release in vibro-compacted hyperstoichiometric mixed carbide of about 5% carbon equivalent.

Swelling

The swelling of fuel must be expected from two sources, namely, from solid fission products and from retained gaseous fission products: in the latter case, the gas may be fixed interstitially in the lattice as atoms, or it may appear as bubbles.

Theoretical estimates of solid swelling have been made by a number of workers, and in UO_2 values between 1.0% and 1.6% volume swelling per

1 at. % burnup have been calculated. The work of Bleiberg *et al.*⁽⁶⁾ with oxide ceramic fuels leads to a total swelling value of 1.7 vol. % per 1 at. % burnup with maximum fuel temperature in platelets up to 2100°C. Results for uranium carbide taken to up to 5.4% at $K(625-900^{\circ}\text{C})$ values of 62 and 95 W/cm by Crane and Gordon⁽⁷⁾ show volume swelling of 2-3% per per cent burnup for stoichiometric or hyperstoichiometric fuel. The results published by Strasser *et al.*⁽⁴⁾ for mixed carbide give immersion density changes of 1-2% per 1 at. % burnup at low burnup and 1.1-1.7% per 1 at. % burnup at about 10% burnup.

Fuel-clad compatibility

No direct compatibility problems between stainless steel and nickel base cladding with oxide fuels at temperatures of interest (up to 700°C) are known, although indirect problems associated with fission products may occur. The case of carbide fuel is dealt with in some detail under evaluation. The main potential problem is believed to be concerned with degradation of the mechanical properties of the cladding due to carbon transfer, and again with suitable reservations, mixed carbide is believed to be compatible with the clad materials considered up to 700°C.

Condensable fission products

In 1% burnup of heavy atoms, each unit volume of mixed oxide fuel produces about 0.004 units of condensable fission products. Thus at 10% burnup, an 80% dense fuel column 100 cm long produces enough condensables (mainly caesium) to fill 3.4 cm of empty fuel pin if it all collected in one place. It is not known in detail to what extent this migration and condensation will occur, nor to what extent solid compounds of caesium will be formed. U.K. workers have found that significant radial migration can occur, with axial migration less pronounced.

Plutonium migration

This could have both physics and metallurgical implications on fuel element performance. On the PFR programme evidence of very substantial plutonium migration (down to 50% of initial concentration) has been found in mixed oxide pins with a $K(650^{\circ}\text{C})$ value of 60 W/cm. The information is not yet sufficiently systematic to incorporate as a design equation.

Strasser *et al.*⁽⁴⁾ have concluded that under performance conditions relevant to fast reactor fuel elements, plutonium migration at 10% burnup is small in hyperstoichiometric mixed carbide.

Cladding

Creep strength, yield strength, ductility, and corrosion resistance in clad materials are the major requisites. In a fast reactor, cladding will typically

be taken to 10^{23} – 10^{24} nvt greater than 0.1 MeV at temperatures in the range 400–700°C in durations of about 15,000 hours. At temperatures below 650°C, radiation hardening due to point defects will occur in both 316 stainless steel and PE 16; at higher temperatures it has been demonstrated by Nettley *et al.*⁽²⁾ that significant loss of high temperature tensile ductility may occur. The effects of a high fast flux on creep rates are virtually unexplored. Some

TABLE 1

	316 L (solution treated)	316 L (cold worked)	PE 16 (solution treated and aged)
Yield stress (10^3 lb/in ²)			
500°C unirradiated	18	57	61
irradiated	~ 80	95	> 100
650°C unirradiated	18	47	65
irradiated	22	45	70
Tensile ductility %			
500°C unirradiated	47	28	41
irradiated	38	10	32
650°C unirradiated	45	35	36
irradiated	32	16	10
Creep stress (1% strain in 10,000 hours at 650°C: 10^3 lb/in ²)	6	20	42

data on post irradiation rupture (Gusev *et al.*,⁽⁸⁾ Nettley *et al.*⁽²⁾) show that irradiation undoubtedly lowers rupture strength in stainless steel and nickel base alloys; a factor of two or three reduction in the 650°C rupture life for stainless steel after irradiation to about 10^{22} nvt has been found.

Table 1 has been summarized from results obtained by U.K. workers on the out-pile creep properties and post irradiation tensile properties of cladding materials of interest: the values are characteristic and subject to considerable variation. The irradiated results are for about 3×10^{22} nvt (fast).

Significant section loss will occur in these materials at the hotter end of PFR pins due to coolant corrosion. Typical figures are given later under "evaluation" and in detail in ref. (2). These authors and Russian workers⁽⁹⁾ find that less corrosion occurs at about 700°C in nickel-base alloys with lower nickel contents, and that the best of them is comparable with austenitic stainless steels as regards section loss.

3. Fuel Element Design Conditions

It has been established elsewhere that an economic fast reactor must employ fuel rated at about 200 W/g (fuel) and taken to between 5% and 10% average burnup of heavy atoms. The design framework within which a successful design must be sought involves the following facts summarized from above: it is assumed that a maximum K value of some 40 W/cm in pins is involved.

Fuel

(a) In mixed oxide, gas release of about 90% will occur in vibro fuel and could be about 50% in pellet fuel. Approximately 25 cm³ (NTP) of fission gas per centimetre of fuel could be released in 10% burnup.

(b) In carbide fuel, a few per cent gas release will occur at up to 5% burnup, but by 10% burnup some 50% of the fission gas may be released. No information on further release on power changing or transients is known.

(c) Fuel swelling of about 2 vol.% per 1 at.% burnup will occur in both oxide and carbide fuel.

(d) Condensible fission products corresponding to 3.4 cm of pin will be formed by 100 cm of 80% dense mixed oxide fuel in 10 at.% burnup.

Cladding

(The remarks below apply broadly to stainless steel and nickel base cladding.)

(a) Irradiation substantially increases the yield stress of solution treated stainless steel below 650°C and reduces its tensile ductility. Above 650°C the yield stress may be lowered and the tensile ductility will be increasingly degraded with increasing temperature.

(b) In cold worked stainless steel, the effect of irradiation is to lower ductility and increase yield strength below 650°C; above 650°C the yield strength is not much altered but ductility is increasingly reduced with increasing temperature.

(c) In solution treated and aged precipitation hardened nickel base alloys, there is a loss of tensile ductility at all temperatures. At 700°C and above the ductility may be very small.

(d) From all the data available, it would be consistent to find cladding failures at not much more than 1% strain in pins.

4. PFR Fuel Element Design

The PFR fuel element development programme has been focused on mixed oxide fuel in a sealed pin design for the first charge. The selection of

oxide instead of carbide was made in late 1962 on the grounds of available performance data, uncertainties of fuel-cladding compatibility with carbide, and the greater manufacturing problems associated with carbide. Sealed pins, with a plenum at the cool end for accommodation of fission gases, were chosen to avoid the substantially unknown reactor circuit problems posed by vented fuel. The pins are located by grids in a hexagonal wrapper. Grid location was chosen for a variety of reasons; alternative designs with fin spacing have been considered, and have been adopted for the radial breeder elements. Table 2 gives parameters relevant to a likely PFR core.

TABLE 2. PFR CORE PARAMETERS

<i>Fuel</i>	
Inner core, $(\text{Pu}_{0.19}\text{U}_{0.81})\text{O}_2$	
Outer core, $(\text{Pu}_{0.25}\text{U}_{0.75})\text{O}_2$	
Effective density	80%
Mean rating of oxide	143 W/g
Maximum rating of oxide	210 W/g
Active fuel length	36 in. (91.4 cm)
Upper and lower breeder length (UO_2)	9 in. (22.9 cm)
<i>Cladding</i>	
Type 316 stainless steel	
Cladding internal diameter	0.200 in. (0.508 cm)
Cladding external diameter	0.230 in. (0.584 cm)
<i>Subassembly</i>	
Hexagonal form, pitch	5.7 in. (14.5 cm)
Pin length	9 ft (2.74 m)
Subassembly length	12.5 ft (3.81 m)
Pins per subassembly	325
Pin pitch	0.290 in. (0.737 cm)
Number of core subassemblies	78
Number of radial breeder subassemblies	42
Core centre mean sodium velocity	20 ft/s (6.1 m/s)
Ratio fuel/stainless steel/sodium in core	0.33/0.20/0.47
Pin spacing	Grids
Weight core subassembly	420 lb (191 kg)
Weight breeder subassembly	600 lb (272 kg)

Pin Design Criteria

The major phenomena which need to be taken into account in designing the pin are fission gas pressure, fuel swelling, thermal stressing due to internal heating and coolant temperature inhomogeneities, and loss of cladding due to corrosion.

Internal fission gas pressure

Fission gas pressure will exert a creep stressing in the cladding which, in isolation, will be of greatest consequence at the (hotter) outlet from the

fuelled section of the pin. From the gas release data discussed by Lawton *et al.*⁽¹⁾ it will be seen that it is a reasonable design assumption to say that all the fission gas is released immediately from oxide fuel at fast reactor ratings, giving a linear rise in pressure. It is found that the primary creep of relevant cladding materials is negligible under these conditions and that the secondary creep can be described by a law of the kind $\dot{\epsilon} = A\sigma^n f(T)$, where A and n are constants, $\dot{\epsilon}$ is creep rate, σ is stress, T is °K, and f is a functional relation. In practice, values of n between 4 and 20 have been found, depending on the material and its manufacturing treatment. From the assumption of linear stress rise, it follows that cladding strain is proportional to time to the power $n+1$, that is, over the majority of the pin dwell time, the strain is very small and then it builds up very rapidly.

The PFR pin was originally designed with the proviso that the creep strain should be limited to 0.1%. It is now believed that this value could be amended upwards somewhat, but in view of the creep law, the permissible dwell time would not be very largely affected. It is perhaps worth noting that this criterion is important in the sense that pins will inevitably fail due to internal pressure if they are left in the reactor long enough and prior failure due to some other mechanism does not occur.

Thermal stressing

The maximum thermal stress induced in a stress-free can at start-up is approximately given by a formula of the kind

$$\sigma_T = \frac{Awr t}{k}$$

where σ_T is stress, A is a constant (related to elastic constants), k is thermal conductivity, w is the fuel rating, r is pin radius, and t clad thickness. Two consequences of thermal stress are considered relevant. First, power changes could lead to Coffin type plastic fatigue of the outer tubing fibres. Secondly, if thermal stress is superposed on internal pressure, incremental increases in clad diameter could occur on power cycling; this has been called "progressive ratcheting". If relaxation occurs, the relevant criterion for an otherwise acceptable design to prevent ratcheting is $\sigma_p + \sigma_T \leq \sigma_Y$, where σ is stress, and suffixes p , T , and Y refer to pressure, thermal, and yield stresses.

Fuel swelling

At present, no adequate theory to take account of swelling is known. Theories making assumptions about the creep and yield properties of ceramic fuel under irradiation can be set up, but shortage of the physical properties data preclude the establishment of working design rules at present. The PFR design has been formulated on the basis that local voidage must be available

to accommodate solid fission product swelling. This leads to a fuel density specification of about 80%.

In implementing the above criteria, due allowance must be made for section loss in cladding by coolant corrosion, and the effect of tubing flaws falling within the specification.

Subassembly Design Criteria

The two main criteria which have conditioned PFR subassembly design are first, that the restraint of grids shall prevent unacceptable pin bowing from occurring, and secondly that the wrapper shall be strong enough to withstand differential pressure arising from the progressive pressure drop on the inside and (effectively) stagnant coolant on the outside, i.e. a uniform external pressure.

Assemblies of highly rated pins have a strong tendency to thermal bowing instability, since any pin deflexion tends to produce a temperature distribution leading to further deflexion in the same direction. This can lead to unacceptable hot spot temperatures before instability conditions occur. Fuel pin bowing is a function of linear heat rating, pin diameter, pin pitch, coolant mass flow, pin to coolant heat transfer, initial bow and deviation from nominal pitch, cladding thermal expansion coefficient, and grid pitch. Theoretical methods have been developed which are believed to predict grid spacing to give acceptable hot spot temperatures.

Bowing of pins imposes stresses in the pins which must be analysed as a function of distance from the free or fixed end of the pin. These stresses must be taken into account as another possible source of ratcheting straining, since in severe cases, plastic hinges may form at grids.

The PFR wrapper is sealed into the core support grid at the lower, cool end. It has been designed on elastic theory, and taking account of the hexagon vertex radius, so that nowhere does the stress exceed the yield stress.

5. Potential Performance Problems

Brief mention is made in this section of possible defect mechanisms other than those covered by the design criteria.

The ratio of plenum to fuel volumes is designed to limit the internal pressure to a value at which the creep strain in the can will be small. "Small" in this context means within the likely ductility of the irradiated cladding, and of such a magnitude that significant loss of coolant flow will not occur. Two effects could undermine the design. First, the pin is designed on out-pile creep data which indicate that primary creep is negligible and that secondary creep will describe the full life of pins. A pronounced irradiation degradation of creep strength would vitiate or invalidate the design basis.

A calculation for solution treated stainless steel indicates that a factor 3 reduction in creep strength (which is thought to be relevant in this case) would reduce the dwell time for a given strain to 80%, so it can be seen that this effect is not dominant. The second effect is blockage of the plenum by condensable fission products. This would be very serious indeed, leading to a rapid build-up of internal pressure and clad rupture. Examination of pins at about 5% burnup gives promise that this latter effect will not be insuperable.

Plutonium migration radially within a pin is not considered to be of much consequence. Axial migration may have some implications in terms of hot spots but the main effect would be on reactivity. Migration of plutonium sufficient to affect the stoichiometry of fuel could have marked effects on the irradiation behaviour of fuel and on cladding/fuel compatibility.

In various fuel element development programmes, reports have been made of internal attack on the cladding. This attack has been variously associated with fission products, fuel impurities and cracks in the fuel. Clearly an internally pressurized pin concept is more at risk from cladding attack than one with a net external pressure.

The PFR design currently employs grids which take note of the fact that axial differential expansion between the pins and the wrapper (to which the grids are effectively attached) could be accommodated by purely elastic deflexion of the grids, together with minor slipping increments between pins and grids. If significant plastic elongations of all the pins occurred, such as might result from fuel swelling, then the elastic grid concept would break down unless pins could all slide simultaneously through the grids. Clearly a number of design solutions to this possible problem are available, including floating grids, and greater pin-grid clearances.

Initially, pin spacing is prescribed to obtain reasonable uniformity of outlet coolant temperature and to minimize hot spots. Any tendency to deposition from the coolant, coolant borne gas hold-up, pin swelling or hold-up of escaped fuel will modify this situation. Detailed studies have shown that very severe local coolant blockage can occur with only a small indication on mixed mean outlet temperature from a subassembly. This is perhaps potentially one of the most serious types of deterioration which may occur, first because of the difficulty of detection, and secondly because of the difficulty of predicting the course and time scale of the sequel.

6. Fuel Manufacturing Development

The main emphasis of the fuel development programme has been on mixed uranium-plutonium oxide fuel which is the reference fuel for the first PFR charge. At the same time, a possible trend to replace oxide by carbide has been taken into account in flowsheeting plant for oxide manufacture. The basic feed materials which could be made available were uranyl nitrate and

plutonium nitrate. The three major considerations which conditioned the choice of manufacturing route were:

- (a) what was the most convenient starting material from overall considerations, nitrate solution or a solid (e.g. plutonium dioxide): considerations of existing plant, forward total reactor programmes, storage, and transport problems are germane to this choice;
- (b) which of the available routes best satisfied physics specifications for homogeneity of the fuel; and
- (c) which route led to the simplest reprocessing route consistent with the optimum usage of available technology and plants.

The choice made is dependent on the particular circumstances of a manufacturer as well as on purely technical grounds. U.K. experience, plant availability, the view taken of meeting physics requirements and the desirability of avoiding hydrofluoric acid in reprocessing (with associated problems of increased γ -active waste storage), led to the choice of co-precipitation of mixed oxide from uranyl-plutonium nitrate solution by ammonia, with filtration and drying of the ADU-plutonium hydroxide precipitate and calcination and reduction to $(\text{Pu,U})\text{O}_2$. Mixed carbide is produced by a further step of carbothermic reduction of the oxide. Physical mixing of the oxides is regarded as perfectly feasible and may well be adopted in Britain eventually.

Two fuel forms have been considered, namely, annular pellets and vibro fuel. Since there are significant differences between the oxide and carbide routes, these will be discussed separately.

Mixed Oxide

Two routes have been considered for processing from the feed powder, one involving the use of a binder, the other binderless. The binder route closely follows established practice for UO_2 manufacture, the essential stages being ball milling, binder addition (e.g. Cranco in CCl_4), and granulation for obtaining a free-flowing material. This feed can then either be pressed into pellets or tableted, fragmented, and classified for vibrofeed, or immediately classified for vibrofeed. Finally, debonding by distillation with a flowing carrier gas (CO_2 , H_2 , or A) and sintering in hydrogen, hydrogen-argon, or carbon dioxide at about 1600°C to achieve the specified stoichiometry completes the route. A binderless route can produce either pellets or vibrofeed, essentially with the binder addition and debonding stages deleted. A typical route would tablet feed powder, the tablets would be fragmented, and either proceed to a classification stage for vibrofeed or go forward as pressfeed for pelleting.

In mixed oxide, both hypo- and hyperstoichiometry can occur, depending

on whether the sintering atmosphere has a reducing or oxidizing potential. A consideration of the fission product spectrum from $[\text{Pu}, \text{U (natural)}]\text{O}_{2.x}$ suggest that the oxide formers amongst the fission products may not be adequate to prevent oxygen liberation and clad attack. This would favour the use of hypostoichiometric fuel. Consequently the development studies have covered specifications from 1.93 to 2.10 oxygen/metal ratio.

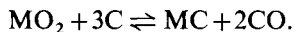
There are sharp differences between the physical characteristics of ceramic grade UO_2 prepared from ADU, and of co-precipitated mixed oxide. For example, whereas the surface area of UO_2 powder is commonly $2\text{--}4 \text{ m}^2/\text{g}$, in mixed oxide, values between 10 and $30 \text{ m}^2/\text{g}$ are found. A conditioning stage for mixed oxide powder is needed to reduce the surface area to give satisfactory granulation.

The sintering of mixed oxides is conditioned by the requirements of specified density and stoichiometry and by impurity levels (particularly of iron). Process specifications have been established which enable fuel with oxygen/metal ratios between 1.93 and 2.10 and densities up to 96% of theoretical to be achieved.

It is remarked that special problems are highlighted in fast reactor fuels due to the presence of α - and γ -activity and neutrons. These pose shielding and binder degradation problems, none of which has been found to present unacceptable difficulties.

Mixed Carbide

Two routes have been examined for the carbothermic reduction of mixed oxide with graphite, namely, vacuum reduction and argon sweep reduction. The reaction proceeds at about 1500°C according to the relation



The maintenance of a low partial pressure of Carbon monoxide is necessary to enable the reaction to proceed to completion. In the vacuum method, pellets of the intimately mixed powders are maintained at $5\text{--}50 \mu$ pressure, giving carbide with up to 0.3% oxygen. The reaction proceeds more completely at higher overpressures if the temperature is raised, but plutonium volatility becomes increasingly more serious: values up to 2% loss have been found.

The argon sweep route effectively keeps down the partial pressure of carbon monoxide by carrying it away as it is released. The choice between the routes is considered to be dominated by convenience of plant layout and operation.

The product of the carbothermic reduction is a hard, porous "coke", which breaks down to form an extremely hard and abrasive powder. This powder is very reactive, taking up oxygen readily from moisture and air-contaminated atmospheres. It has been found necessary to specify argon

with a limit of 25 ppm (O + H₂O) for handling this powder in order to meet an acceptable final product specification.

In the onwards fabrication of carbide, it has been found to be necessary to incorporate a binder to deal with impurity pick-up, the possibility of pyrophoricity, and marked tool wear with dry feed. Paraffin wax, either alone or borne in a suitable carrier, has been found to be effective; nickel additions are made at this stage to promote subsequent sintering. Sintering, at about 1500°C, can be carried out either in hydrogen, in which case a near-stoichiometric product results from the reaction $M_2C_3 + 2H_2 \rightarrow 2MC + CH_4$, or alternatively in a non-reducing atmosphere, in which case final control of carbon and oxygen content must be achieved by careful regulation of the feed material. Apart from process detail, fabrication of carbide pellets or vibrofeed is similar to that for oxide.

Pin Filling

The loading of pellets is straightforward. There is a requirement to specify can-pellet clearance which may result in the need to grind pellets. This is a relatively simple process either with oxide or carbide. With vibrofilled pins, broadly two techniques can be used. The first employs the simultaneous loading of graded granules with a high energy compaction phase of greater or lesser complexity depending on the final density specification. The alternative process, developed for PFR fuel, is to load successively finer fractions by an infiltration technique. In principle, the feed can be angular (as from Dynapak, crushed pellets, etc.) or spheroidized.

For PFR, where the filled density specification has varied around 80%, the infiltration technique has been found to be very satisfactory. The packing is essentially controlled by particle geometry, i.e. tubing diameter variations are reflected in varying weight for a given length of fuel column. It has been found that density variations within $\pm 2\%$ of the mean can be achieved by either technique, with a local end variation up to about 5%.

7. Fuel Element Manufacturing Development

Cladding

The development of reactor grade cladding involves arriving at a close, but practicable, impurity and cleanliness specification, specifying and obtaining adequate control of dimensions and mechanical properties, and laying down workable inspection requirements. In all of these specifications there is a degree of arbitrariness which can only be resolved initially by experience and ultimately be reduced by lengthy performance studies. A balance must always be struck between perfection and cost. In the PFR case, tube wall thickness in the range 0.015–0.025 in. (0.038–0.064 cm) has been examined. The working rule has been adopted that imperfections, inclusions, and

average grain-size should be confined to about 10% of the wall thickness, in the absence of special pleading to the contrary. The most contentious area has been in the establishment of standards for, and inspection of, defects. For this purpose, ultrasonic inspection methods have been developed which are capable of obtaining equal resolution of bore and surface defects 10% wall thickness $\times 0.050$ in. (0.130 cm) perpendicular to the tube surfaces and either longitudinal or transverse. Initially it is proposed to inspect all tubing to this standard. Experience will show whether this is essential and whether a sampling technique can be adopted satisfactorily.

At present, high quality stainless steel tubing can be obtained meeting a tight specification. This is not yet the case with nickel base alloys, which are less highly developed, but no intractable problems are recognized.

In the PFR design, the tubing is drawn down to a minor diameter over the plenum length to reduce pressure drop. This complicates the empty can route, but is a practical proposition.

Pin Details

The fabrication of pins involves fuel loading, location of fuel, gas filling and end sealing. For PFR either vibro or pellet fuel may be adopted, involving some differences of fabrication route. Fuel loading has already been mentioned. Two considerations in the location of fuel arise. First, a datum must be established in the pin for controlling the position of the column as loaded, and therefore the core reactivity worth; secondly, the PFR design has compartments within the fuel column both to restrict redistribution, and to locate the fuel segment-wise with respect to the can. This latter requirement is of a somewhat conjectural nature, but may result in a very material lessening of peak temperature in a transient resulting from pump run-down.

Two details have been studied for locating fuel, namely, features brazed to the can and features crimped to the can. The brazing method involved the study of braze compatibility, braze flow control, and the effect of contaminating atmospheres. This work has been carried to the stage where an acceptable process and product can be recommended for oxide pins. The performance objection to brazes has been the attendant metallurgical modification of the canning material. A programme has therefore been carried out on the mechanical location of platforms in pins by pressurizing into a groove in a pellet. This has covered process studies, and evaluation of the strain concentration in the indented area and the degree of fixity of the platform. No choice has yet been made between these methods, but neither has so far been ruled out from performance considerations.

End sealing of pins inevitably brings into consideration the optimum cladding, metallurgical condition recommended, and problems of assembling pins through grids. In addition, it is conjectured that the cladding of fast

reactor elements will be so reduced in ductility at high burnup that strain concentrating features should be avoided if possible. The philosophy has been adopted for PFR that failure should not occur in a closure before it would occur in the body of the tube. This has generated a programme of investigation of brazing, welding, and forming closures in clads of varying heat treatments. Another requirement is that the process adopted should enable manufacture to proceed essentially on a process control basis rather than by refined inspection of the product. In these programmes due note has been taken of the possible need to change the cladding dimensions. Evaluation of the process is seen in two stages. A broad sorting programme on unirradiated specimens has been carried out in rapid bursting and rupture type tests. Presently in hand are irradiation trials both on pins and specimens.

Welding studies showed that acceptable quality and reject rates required the development of a special purpose welding machine⁽¹⁰⁾ with electronic timing control of rate of heat input, and tape control of the welding sequence.

Assembling

The problems posed by subassembly manufacture are the manufacture of shroud tubes and grids to specification, the location of grids within the wrapper, and the ability to load pins without damage. In the PFR case, the specification for wrappers involves the across flats size locally, with twist and bow limited so that the wrapper lies within a hexagonal envelope: this specification has not yet been achieved in practice. High quality grids have been obtained to meet a pin pitch specification of ± 0.005 in. (0.013 cm). The problem of pin assembly is complicated by the consideration that eventually γ and neutron activity of recycled plutonium will require pins to be loaded remotely behind shielding. Two approaches are under review for PFR. The attachment of grids to the wrapper minimizes the problems of control of edge-pin coolant gaps, but means that possible pin damage during loading may not be detected. The assembly of the pins into an inspectable grid cage with the bundle subsequently inserted into a wrapper complicates control of edge coolant. The PFR approach is to tackle the former method in an attempt to minimize hot spot allowances.

8. Evaluation Results

Pin Design Detail

The evaluation programme aimed at selecting end closure and pin internal design detail has concentrated on a 316-type steel. This has been used both in the solution treated condition (1050°C, 20 min) and with varying degrees of cold work, mainly around 20% reduction of area. Typical results are quoted in Table 3 for rapid testing at room temperature and at 650°C.

With solution treated material, it can be seen that the forms of end closure

studied do not nucleate failure at the end. With cold worked material, the design of end closure has to be developed to prevent the end closure from impairing the inherent potential of the tube.

TABLE 3. RAPID BURSTING PRESSURE (10^3 lb/in²)

	20°C	650°C
316, solution treated	10	6
316, 20% cold worked	18	10
Swage end closure, S.T. tube	Failed in tube	
Swage end closure, C.W. tube	Failed in tube	
Butt welded endcap, S.T. tube	Failed in tube	
Butt welded endcap, C.W. tube	14	8
Brazed fuel spacer, S.T. tube	Failed in tube	
Brazed fuel spacer, C.W. tube	14	8
Crimped fuel spacer, S.T. tube	Failed in tube	
Crimped fuel spacer, C.W. tube	Failed in tube	

The fixing of intermediate spacers follows a similar pattern. Brazing or crimping do not provide a preferred site for failure in solution treated tubing, but brazing causes failure in cold worked tube near the site of the braze. It is remarked that this position changes with long exposure at temperature. Moreover, the interpretation of the results given above must include an awareness that the data are conditioned by the ductility of unirradiated material, since this will not be relevant to irradiated pins.

Fuel-clad Compatibility

No specific programme has been mounted on compatibility between mixed oxide fuel and cladding. The results obtained from pins irradiated to 700°C do not indicate a general problem although instances of cladding attack have been noted.⁽¹⁾

A substantial out-pile programme at 650°C and 800°C for exposure times up to 800 hours has been carried out with mixed carbide of varying stoichiometry using stainless steel and nickel base cladding. In the majority of cases a helium bond was used, with a sodium bond in only a few cases.

The results may be summarized as below: those for stainless steel have been reported in greater detail.⁽¹¹⁾

	Stainless steel	Nimonic alloys
Hyperstoichiometric fuel 650°C	Slight carbon transfer; no reaction with fuel	Slight carbon transfer; no reaction with fuel
Stoichiometric fuel 650°C	Slight carbon transfer; no reaction with fuel	Slight carbon transfer; no reaction with fuel
Hyperstoichiometric fuel 800°C	Slight carbon transfer; no reaction with fuel	Significant reaction to a depth of 3 mils (0.008 cm)
Stoichiometric fuel 800°C	Slight carbon transfer; no reaction with fuel	Significant reaction to a depth of 3 mils (0.008 cm)

At present, the degree of carbon transfer with stainless steel is not regarded as cause for concern, but this topic needs to be pursued to see whether any complications due to irradiation occur. At 650°C the Nimonic alloys are regarded as promising for further study: 800°C may be above the service limit with mixed carbide fuel.

The presence of sodium very materially increases the rate of carbon transfer from hyperstoichiometric fuel to stainless steel. In this context, it will be appreciated that initially single phase fuel may become two phase under irradiation.

Coolant Corrosion

Two phenomena arise from the sodium coolant, namely, cladding corrosion and carburizing. The effects of oxygen content over the range 5–30 ppm,

TABLE 4. CORROSION LOSS FROM ONE SURFACE

Material	Oxygen (ppm)	Temp. °C	Velocity		Loss	
			ft/s	m/s	(10 ⁻³ in/a)	(10 ⁻³ cm/a)
316	10	650	16	4.9	0.6	1.5
316	10	650	40	12.2	0.7	1.8
316	10	725	16	4.9	1.2	3.0
316	25	650	16	4.9	1.8	4.6
316	25	650	25	7.6	1.8	4.6
316	25	725	16	4.9	4.5	11.4
PE 16	25	650	16	4.9	1.7	4.3
Nimonic 80A	10	650	15	4.6	2.0	5.1
Nimonic 80A	10	650	40	12.2	5.0	12.7

and velocity in the range 5 ft/s to 40 ft/s (1.5 m/s to 12.2 m/s) at temperatures of 650°C and 725°C have been examined. With stainless steel, the rate of loss is sensitive to oxygen content and broadly insensitive to velocity at velocities above 10 ft/s (3.0 m/s). Typical values are given in Table 4. With high nickel content alloys (about 80%), the rate of loss is sensitive to velocity and is regarded as marginally too high for PFR. With intermediate nickel content (PE 16, 43% Ni) the cladding behaves more like stainless steel. From these results, obtained in small scale laboratory tests, it is seen that at temperatures up to 725°C, a simple design allowance should be adequate to deal with corrosion if the oxygen content of the sodium is kept low. It remains to check that the data are valid for fuel elements in rigs more analogous to a reactor circuit in size and construction.

Results of carbon pick-up from both cold-trapped and hot-trapped sodium have been obtained in laboratory rigs by Nettleby *et al.*⁽²⁾ They show that at 650°C, surface carbon in stainless steel can increase from 0.02% to 0.2%

in a few months. The implications of these results on fuel element performance remain to be clarified. Nettlely shows that loss of high temperature ductility must certainly be expected, but at present no very serious degradation of pin cladding performance can be predicted from this phenomenon.

Subassembly Hydraulic and Mechanical Testing

Flow information is required for two design aspects of subassemblies, namely, pressure drop and flow distribution and mixing.

Pressure drop

Pressure drop data are acquired in two stages. The detailed contributions of friction and form loss due to pins and grids are obtained on large-scale models operating on air, while overall loss is measured directly on a prototype using water. The equipment is described briefly in another paper.⁽¹²⁾

The pin frictional loss is a function of the axial spacing between grids. With a pin bundle with effectively no grid support, the hydraulic diameter concept, using conventional pipe friction losses, is accurate to about 15%. With support grids offering 27% blockage to flow area (relevant to DFR experimental subassemblies), and spaced apart by twelve equivalent hydraulic diameters, the frictional loss alone can increase by up to 16%; the total loss with the grids and pins together is increased by about 70%.

The losses calculated for grids using conventional expansion, contraction, and friction coefficients have also been compared with experiment and shown to be some 25% too low.

Measurements across a complete full-size subassembly have shown that the pressure drop can be some 10% higher than that obtained on ideal large-scale models. This is attributed mainly to the effect of departures of the detail of the full-scale prototype from the models.

Flow distribution and mixing

The detailed distribution of flow entering a subassembly combines with the flux to delineate unrelaxed hot channel factors. The turbulent mixing of fluid combined with thermal conduction relaxes the temperature gradients and provides final hot channel data. The techniques of flow distribution and mixing experiments for PFR have been discussed by Eickhoff *et al.*⁽¹²⁾

The distribution of flow between channels defined by three pins, two pins and the wrapper, and one pin and the wrapper corner, have been measured for several minor variations of the basic geometry and for variations within these geometries due to manufacturing tolerances. Measurements have been compared with theoretical estimates, and so far it has not been possible to obtain agreement better than about 20%. The ratio of the centre-to-edge channel flows, for example, has been shown to be very dependent on the

detailed grid geometry. All design changes, other than minor changes, are therefore still tested experimentally.

The enhancement of the fluid mixing, induced by the presence of the grids, has been measured. The degree of enhancement has been shown to be a function of detailed grid geometry, flow blockage area of the grid, and axial spacing of the grids. With particular grids in a DFR experimental subassembly offering 27% blockage and spaced at 2 in. (5.1 cm) intervals, the enhancement is about $\times 5$, giving a mean eddy diffusivity of mass over the 2 in. (5.1 cm) spacing of about $25 \text{ cm}^2/\text{s}$ at a Reynolds' number of 2.25×10^4 . This is equivalent to an effective increase in the thermal conductivity of NaK (appropriate to DFR) of about a factor 2. At PFR Reynolds' numbers, the eddy diffusivity is equivalent to a factor 2 increase in sodium conductivity. The mixing enhancement has been shown to derive entirely from eddy shedding from the downstream edge for grids of any significant depth. Variations in the cross-section of the grid strips from that of a simple rectangle have led to increases in mixing by a factor 10, at the expense of some pressure drop.

Endurance testing

The submission of subassemblies to endurance runs in sodium at reactor temperature and flow, and to thermal shocks representative of power changing, has been regarded as a necessary part of the out-of-pile endorsement of designs. The fuel element thermal test rig and DERE 4 in. (10.2 cm) loop, described in ref. (12), were built for this purpose. These tests endorse the final design from considerations of corrosion and mechanical strength. Thus, while in an early test inadequate standards of pin manufacture were shown up, in a more recent test in which a DFR experimental subassembly was submitted to 1000 hours' steady running in NaK at 600°C and twenty thermal shocks between 600°C and 200°C , nothing untoward was revealed, giving confidence that the design is adequate for reactor loading.

Pin-grid interactions

One detailed aspect of subassembly design which has received consideration is the interaction between pins and grids resulting from relative movement caused by differential thermal expansion. This problem has become more significant as the strength of grids has been reduced, as a result of design changes aimed at reducing pressure losses. Ratcheting experiments in sodium at temperatures of up to 600°C between small groups of pins and grids have been carried out, and load-deflexion measurements on grids in air at room temperature have been made. A theoretical treatment of the problem has been carried out. It has proved possible to design a grid that is sufficiently elastic to move with the pins over the major portion of its area, yet sufficiently strong to withstand coolant forces. Reasonable agreement has been achieved between theory and experiment in this design.

9. Fuel Element Irradiation Programme Results

The PFR irradiation programme has had two definite aims corresponding to two phases of development. The first has been to establish that a burnup target of 5% (average) of heavy atoms at a hot spot temperature of 600°C could be reached: this was regarded as the basic requirement for embarking on a prototype reactor. The second target is to extend the fuel performance to 10% (maximum) burnup at 650°C hot spot. In addition a variety of experiments has been conceived to examine alternative materials and design concepts. It should be emphasized that these targets implicitly refer to designs having a low and acceptable fuel element failure rate: precisely what is meant by acceptable will only emerge from the operation of PFR. The Russian experience with BR 5 exemplifies this point clearly.⁽¹³⁾ They reported the first signs of failure in a particular design at 2% burnup (for which the charge was designed). This charge was then taken to 5% burnup with a subassembly defect incidence of about 20%.

The targets delineated above were set for mixed oxide fuel. The programme mounted in the DFR comprised single and three pin irradiation rigs, and rhomboid subassemblies, 4 × 2 in. (5.1 × 10.2 cm) across the major and minor axes and holding seventy-seven pins of reference size. Details of the results are reported more fully.⁽¹⁾ The salient results obtained to date are summarized below:

- (i) seventy-seven pin subassembly: mixed oxide fuel clad in 316-type steel; av./maximum burnup 4.8%/5.5%; hot spot temperature 590°C. No failures were found in these pins;
- (ii) mixed oxide pins: stainless steel clad; av./maximum burnup to 6.8%/7.7% without failure;
- (iii) failed pins.

A number of pins have been run beyond the nominal design life. This does not imply that failure was expected in all cases, but that increasingly less design conservatism was present. The total of oxide pin failures which has occurred is four. Two of these occurred at 7.1% and 8.1% average burnup in short pins which are believed to have run very substantially above their creep endurance limit. Failure was by cladding splits. A further failure, in a long pin, at 6.1% average burnup was detected by weighing and a gas extraction determination; no clad deterioration was found. The remaining failure at 7.6%/9.1% av./maximum burnup was in a standard annular pellet pin and is attributed to fuel swelling.

A number of experiments are now proceeding to a point where failures are regarded as possible or likely.

10. Design Review

The results of the development programme to date indicate that 5% burnup at up to 650°C clad temperature can be achieved with a suitable design. It is conjectured that fuel swelling constitutes the most immediate bar to reaching 10% burnup. There is no serious doubt that 10% burnup can be reached, but it is a matter for detailed economic study to decide the optimum burnup to aim for, having regard to the design parameters which must be used to achieve a given burnup. Nothing has emerged on the PFR programme to date to give any cause for concern either in the normal or defect behaviour of fuel elements.

This paper has concentrated on the development of mixed oxide fuel in stainless steel cladding with some mention of alternative materials. The range of development being covered in U.K. studies is summarized below.

Fuel:	Mixed oxide Mixed carbide
Cladding:	Stainless steel Nimonic alloys Molybdenum alloys
Design:	Diameter and wall thickness variations Fuel density variations Vented fuel

The role of carbide in a fast reactor programme can only be seen in perspective when its performance limitations relative to oxide have been determined. This will be a relatively extended process. The present U.K. view is that as a simple replacement for mixed oxide, enough is known about carbide to doubt its viability with the reference design in the U.K. programme. If it is found possible effectively to exploit its advantages of gas retention and/or high thermal conductivity, it opens design possibilities which may correspond to cheaper or more durable elements than are possible with oxide. Two obvious avenues are larger diameter pins and thinner cladding.

It is not clear at the moment what is the best balance between strength and ductility in cladding. More fuel element data are required. Molybdenum has been included in a minor way in the U.K. cladding programme as representing a material with potential for a very high temperature operation, which could offset its nuclear drawbacks.

The major problems of gas release and fuel swelling will presumably be influenced by effective changes to cladding and fuel strength. These can be effected by thickness/diameter ratio changes in the clad, variations of cladding material and heat treatment, and fuel density changes.

The role of vented fuel is not immediately clear if fuel swelling is the major bar to high burnup. Clearly it removes the problem of creep of the can

under internal fission gas pressure. Vented fuel is included in the U.K. programme in the expectation that the trend will be to more advanced steam conditions and therefore to higher can temperatures, when creep failures will assume major importance.

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